

PoWER Simulation Update

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① Main Simulation

② Prototype

③ Conclusions

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③ Conclusions

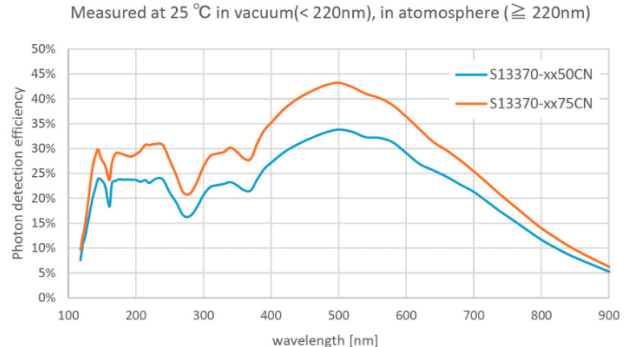
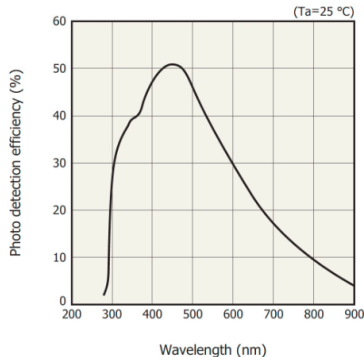
④ References

Main Simulation

Optical Parameter	Value
Photon Yield (500V/m)	25 ph/keV
Xe dopping	10 ppm
Fraction 128 nm	30%
Fraction 175 nm	70%
PEN conversion 128 nm	50%
PEN conversion 175 nm	50%
PEN coverage	FC, 50% cathode
Rayleigh scattering 128 nm	1 m
Rayleigh scattering 175 nm	8.5 m
Absorption length 128 nm	20 m
Absorption length 175 nm	80 m
Reflectivity Anode	30%
Reflectivity ESR	98%

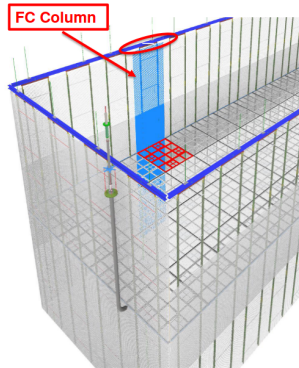
SiPM Efficiencies

From Ana and Francesco's presentation in PoWER kick-off Meeting



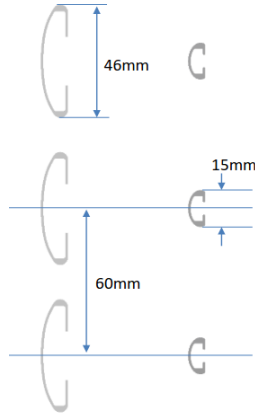
In PEN emission peak (450 nm) the efficiency for visible sensitive SiPMs is about 50%, but for UV sensitive SiPM is about 30 % - 45%.

Field Cage



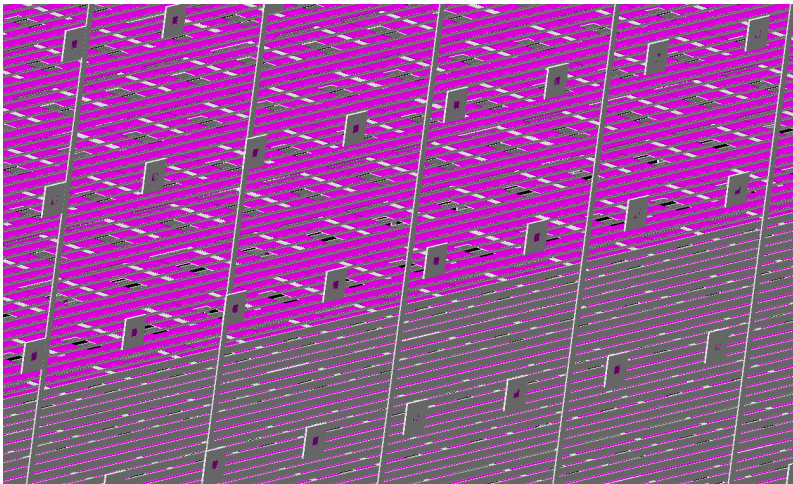
Average Transparency:
~ 45%.

FD1 profile narrow profile

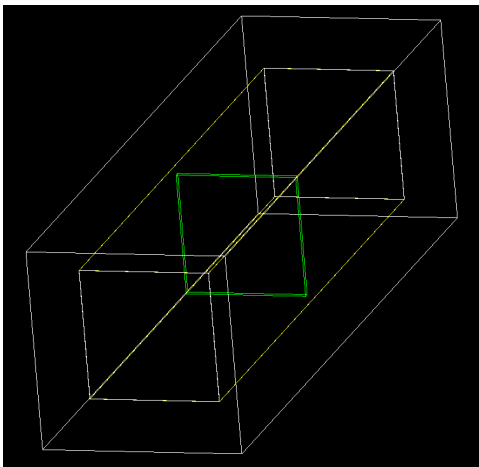


- FD 1 profile
Transparency: ~ 23%.
- Narrow profile
Transparency: ~ 75%.
- Narrow profiles cover the long walls 2.5m from the cathode.
- Vertical bars:
 $13\text{m} \times 45\text{mm} \times 8\text{mm}$;
- Distance between two vertical bars: 1.8m.

Field Cage - New Design

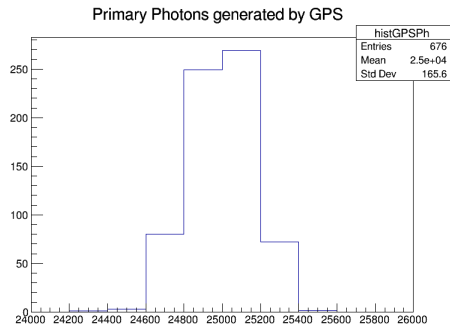
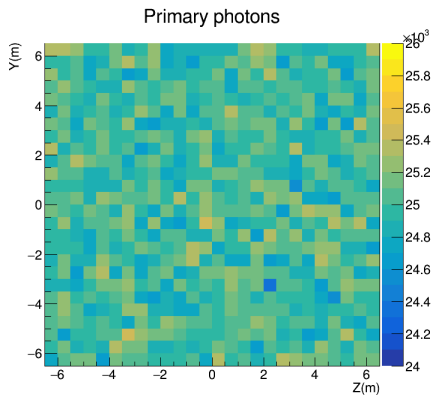


Light Map Study

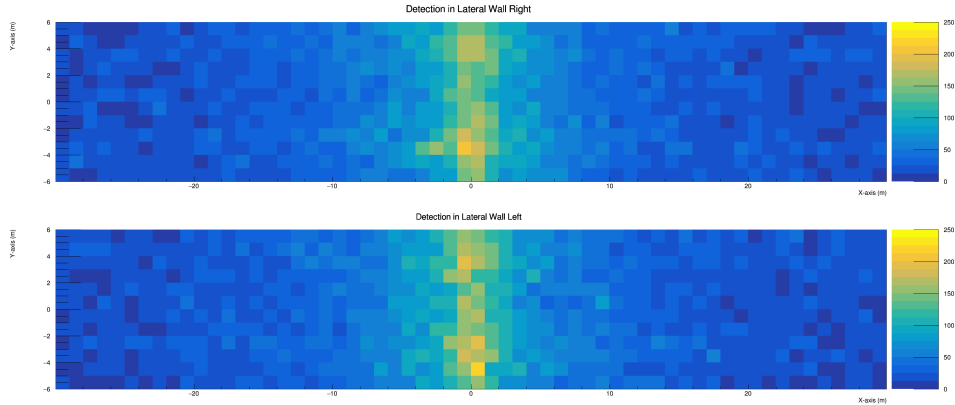


- Photons produced randomly over different positions inside a parallelepiped of dimensions $6\text{m} \times 6\text{m} \times 0.5\text{m}$. The directions and polarizations are random.
- Total of 14.4 million photons, resulting in approximately 25,000 photons/MeV per voxel of dimensions $0.5\text{m} \times 0.5\text{m} \times 0.5\text{m}$.
- For veto studies, we use an extended parallelepiped of $6.5\text{m} \times 6.5\text{m} \times 0.5\text{m}$ and increase the number of photons to 16.9 million photons.

Primary Photons

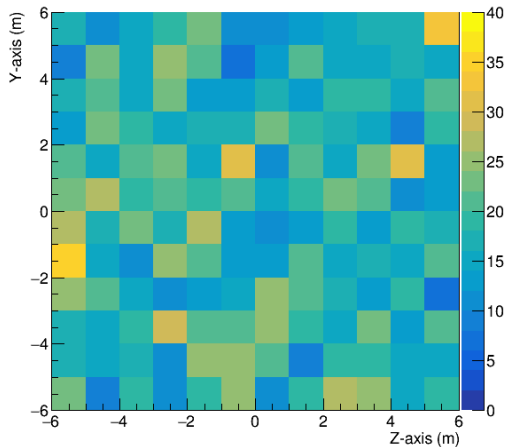


Signal on Detectors

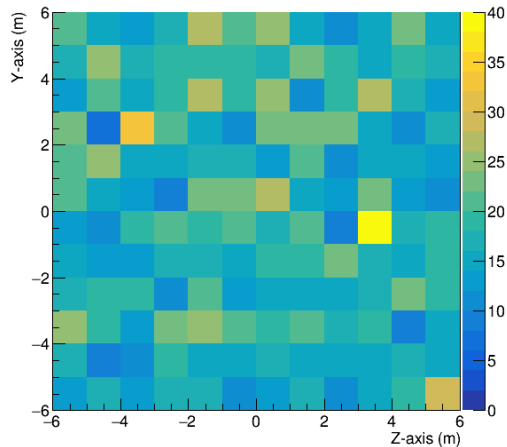


Signal on Detectors

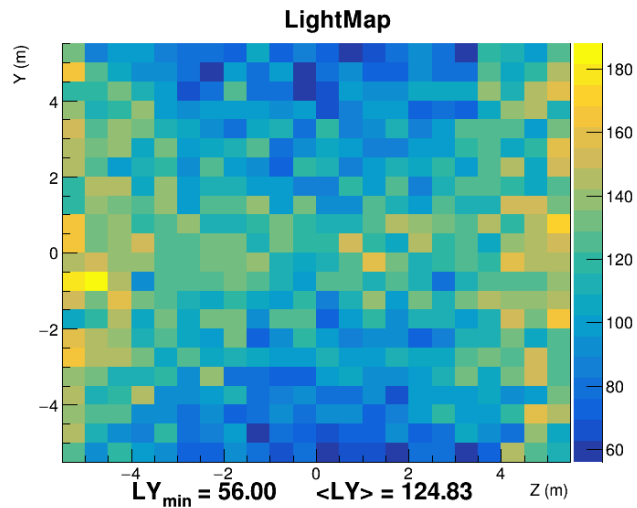
Detection in Front Wall



Detection in Bottom Wall



Current Status - LightMap



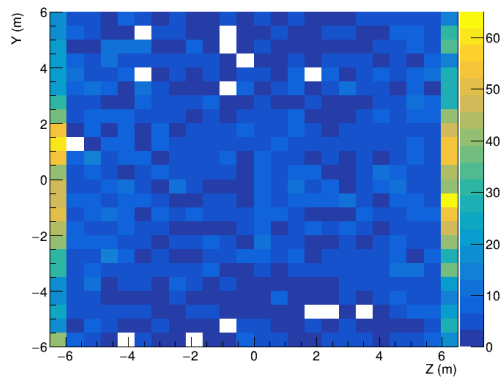
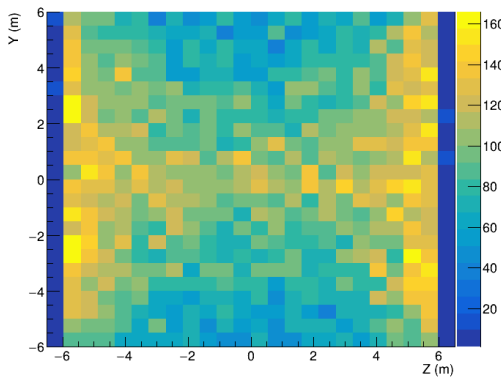
Current Status - Veto

- The idea is to identify events that take place in the buffer zone, outside FC.
- Events occurring inside FC generate photons that are converted to visible by the PEN. They are detected by both detectors.
- Events occurring in the buffer zone also contain unconverted UV photons. They are unable to go inside the field cage as they are blocked by the acrylic foil, and are detected almost exclusively by UV detectors.
- Outside the FC, the photon yield is larger. Nevertheless, the proportion between UV and VUV photons serves as an indicator:

$$R = \frac{LY_{VUV}}{LY_{VUV} + LY_{VIS}}.$$

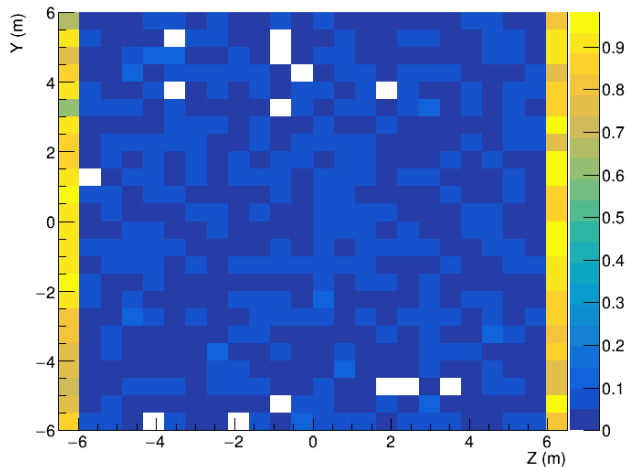
Events occurring inside should have $R \approx 0$, and outside should have $R \approx 1$.

Current Status - Veto



Left: Visible. Right: VUV.

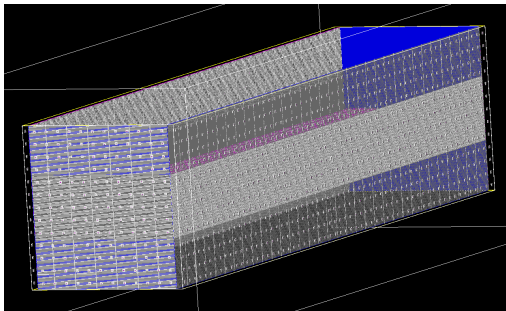
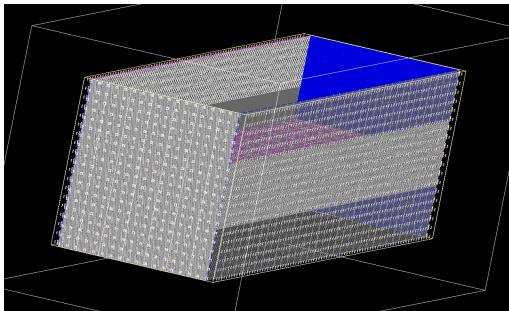
Current Status - Veto



Introduction

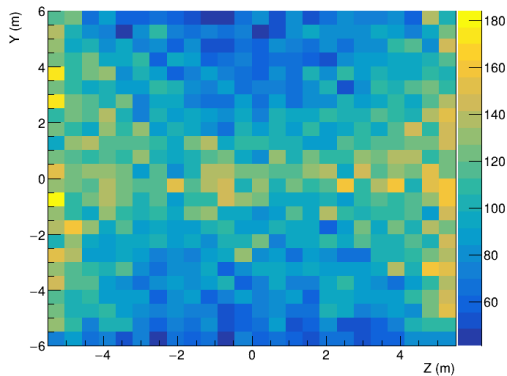
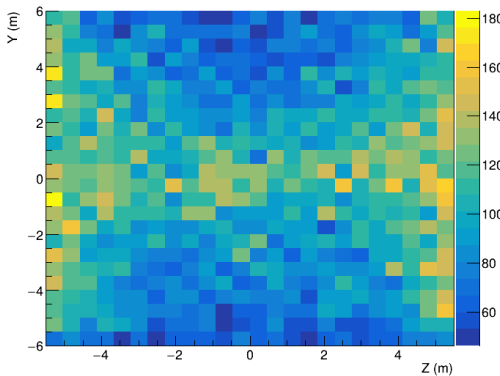
- The main simulation has all required elements implemented.
- The default design provides $\langle LY \rangle = 125$ PE/MeV and $LY_{min} = 56$ PE/MeV. Sufficient for DUNE physics goals, but inferior to APEX.
- The veto system is demonstrated to work in the simulation.
- The focus now is optimizing the LY performance, **without jeopardizing the veto**.
- We use the simulation to test some modifications to the original design:
 - Field Cage modification
 - Number of LDUs
 - PEN efficiency improvement

New Field Cage Proposal



New Field Cage Proposal - Lightmaps

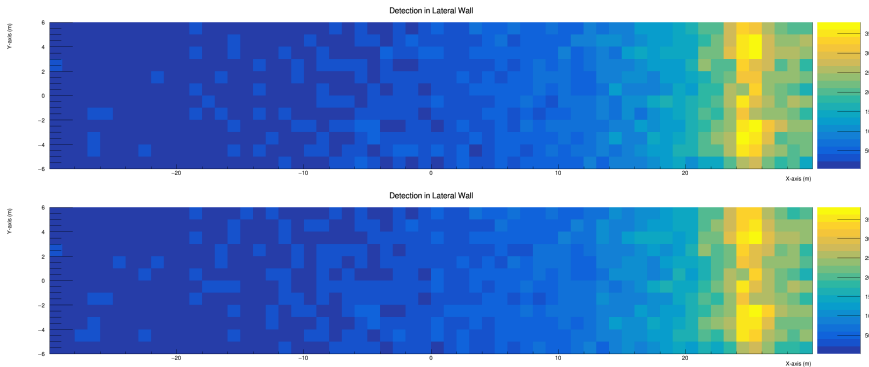
Left: Original FC design. Right: Proposed design.



There were no significant differences in $\langle LY \rangle$ and LY_{min} .

New Field Cage Proposal - Signal

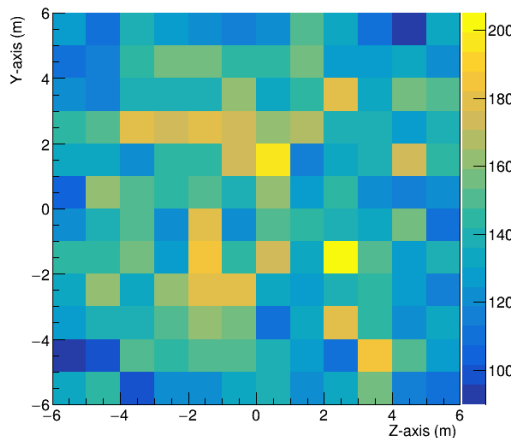
Left: Original FC design. Right: Proposed design.



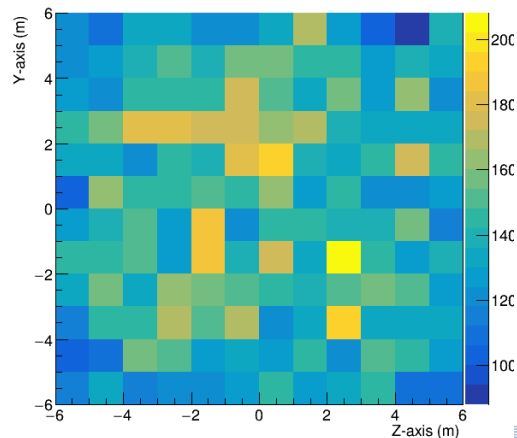
New Field Cage Proposal - Signal

Left: Original FC design. Right: Proposed design.

Detection in Front and Bottom Wall

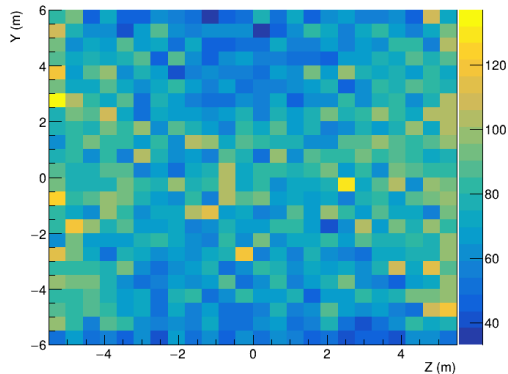
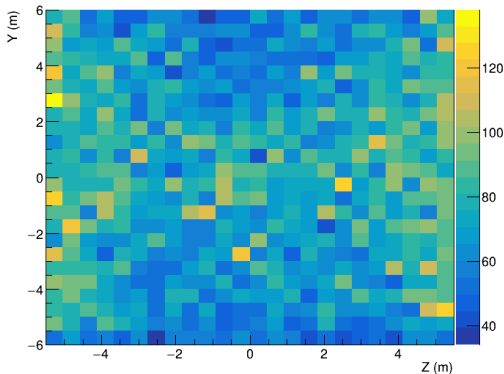


Detection in Front and Bottom Wall



New Field Cage Proposal - Lightmaps

Left: Original FC design. Right: Proposed design.



There is no significant difference when the source is close to the front/bottom walls.

Number of LDUs

Variation of the space between LDUs (and the number of them). LY given in PE/MeV.

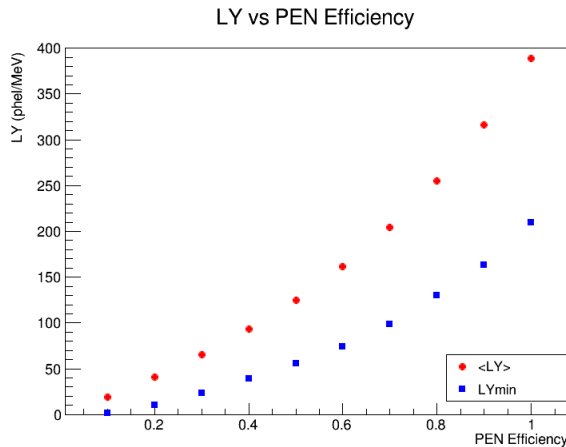
Distance (m)	LDUs	Area (m ²)	<LY>	LYmin
1	1872	75	124.83	56
0.9	2400	96	157.7	64
0.8	3060	122	196.42	86
0.7	3876	155	245.87	120

APEX: <LY>=180, LYmin=109.

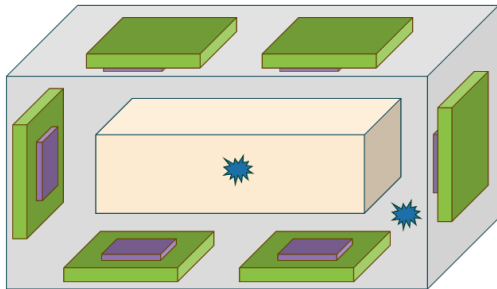
APEX area: ~7000 PD units with an area of 50×50 cm²/unit ~ 1750 m²

F. Marinho arXiv:2503.06032

PEN Efficiency



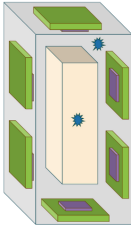
Prototype



- **External Chamber:** Covered with reflector SiPM matrix on the sides.
- **SiPM:** Hamamatsu type:
S14160-6050HS - VIS
S13370-6050CN - VUV
- **Internal Chamber:** PEN or PEN+ACRYLIC
- **Source:** Gamma, Beta, Alpha:
241Am - 133Ba - 57Co - 137Ce -
22Na - 90Sr - 235U

Prototype

LIGHT-YIELD



- 6 SiPM arrays
- Photosensor surface:
 $(2,4\text{cm} \times 2,4\text{cm}) = 5,76 \text{ cm}^2$
- Total active area: $6 \times 5,76 = 34,56 \text{ cm}^2$
- Reflector surface = 810 cm^2
- Coverage = $34,56/810 = 4,3\%$



8x8 S1416 1	8x8 S1416 1	8x8 S1416 1	8x8 S1416 1
8x8 S1416 1	8x8 VUV4	8x8 VUV4	8x8 S1416 1
8x8 S1416 1	8x8 VUV4	8x8 VUV4	8x8 S1416 1
8x8 S1416 1	8x8 S1416 1	8x8 S1416 1	8x8 S1416 1

1	5	9	1
2	6	1	3
3	7	0	4
4	8	1	5

2 0

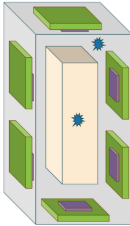


Prototype



LIGHT-YIELD

$$LY = \text{PhotonYeld} \times \text{PEN Eff} \times \text{Collection Eff} \times \text{PDE SiPM}$$



$$\text{Collection Eff} = \text{Coverage} / (1 - R(1 - \text{Coverage}))$$

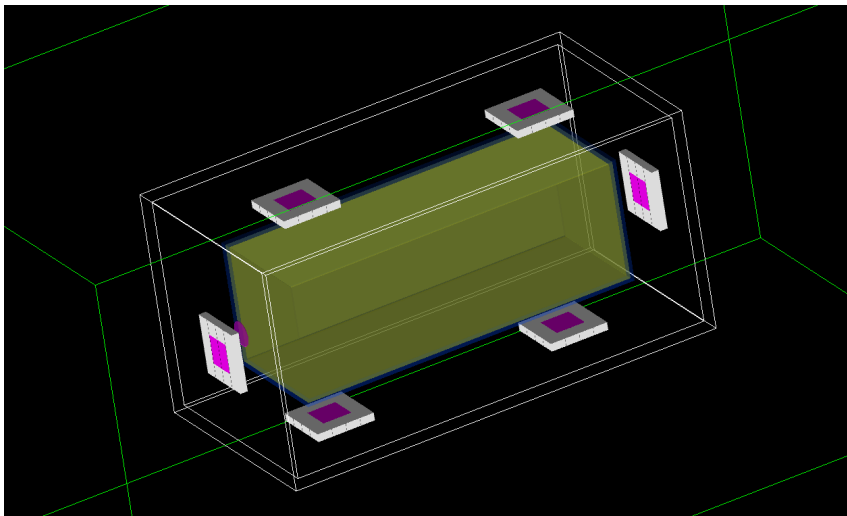
R --> reflectivity of (Reflector + PEN) @ 420nm

$$\text{Collection Eff} = 0,043 / (1 - 0,9(1 - 0,043)) = 0,31$$

$$LY = 40 \times 0,5 \times 0,31 \times 0,5 \approx 3,0 \text{ phel/KeV}$$



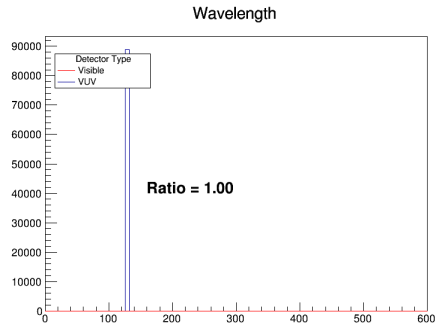
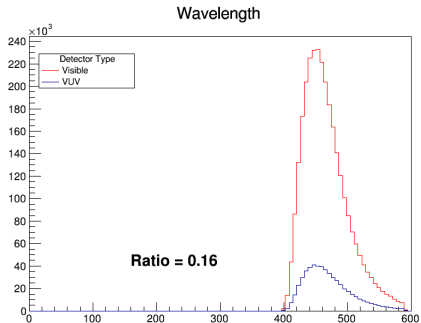
Prototype



Light Simulation

- We use a photon box of dimensions $8\text{ cm} \times 8\text{ cm} \times 16\text{ cm}$
- We generate photons inside and outside the PEN+Acrylic box
- We generate approximately 40000 photons at each voxel of dimensions $1\text{ cm} \times 1\text{ cm} \times 1\text{ cm}$.
- $LY \approx 11\text{ phel/KeV}$ for photons generated inside the PEN+Acrylic box.
- Larger ESR reflectivity used in simulation (0.98) may explain LY.

Light Simulation



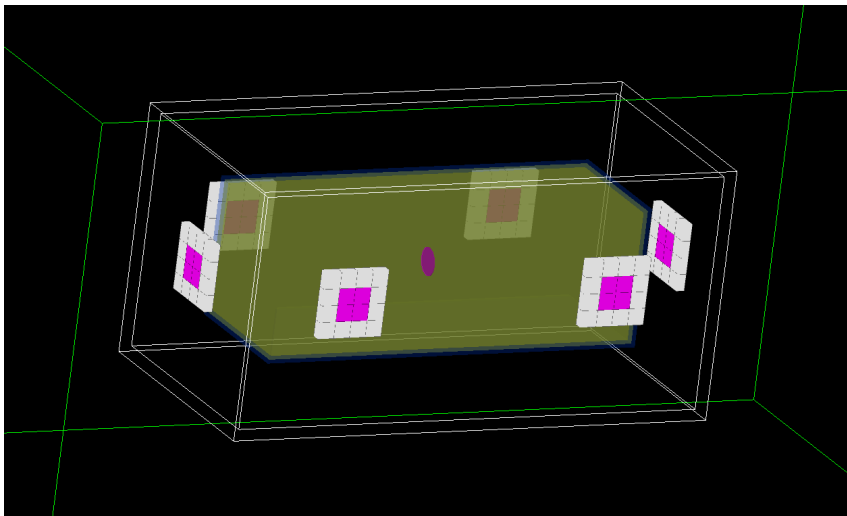
Alpha Source

- Alpha source to be used: Uranium-Aluminium alloy
- Three isotopes presented: U234, U235 and U238
- The source is modeled by generating only the respective alpha particles with respective abundances.

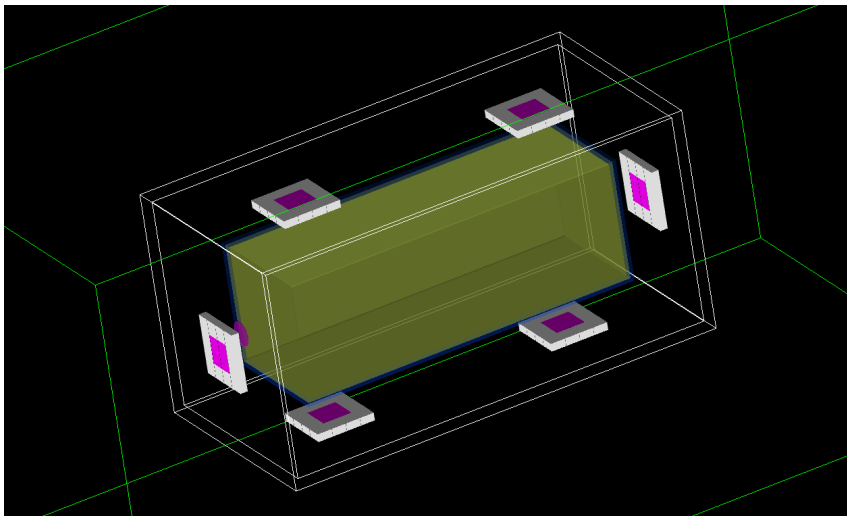
α energy (MeV)	relative intensity	parent nucleus
4.187	48.9%	^{238}U
4.464	2.2%	^{235}U
4.759	48.9%	^{234}U

H.V. Souza et al 2021 JINST 16 P11002

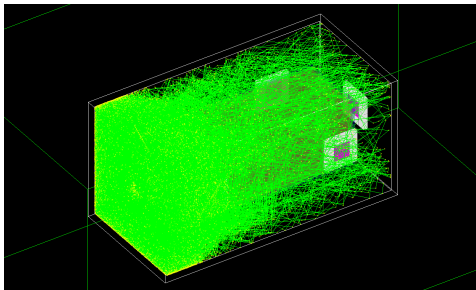
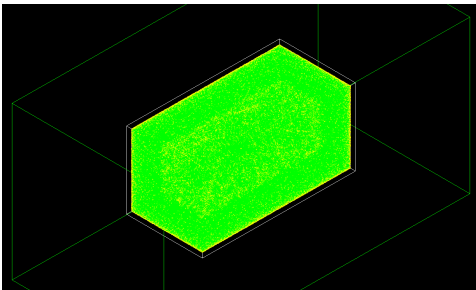
Prototype



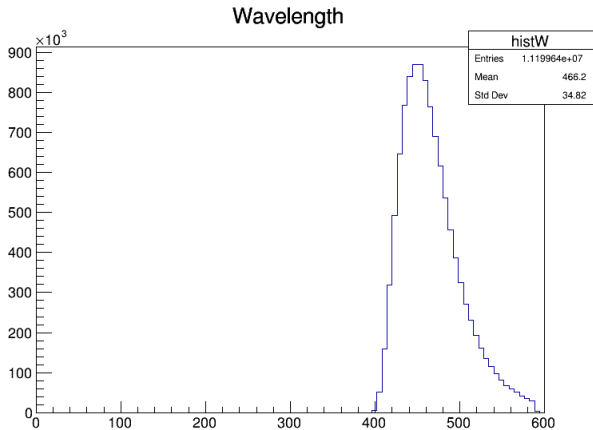
Prototype



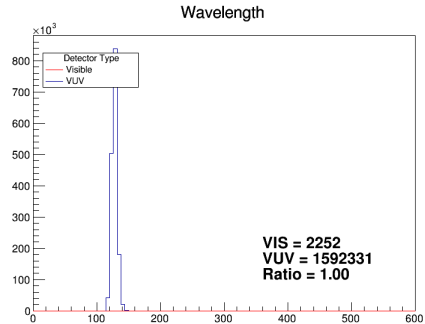
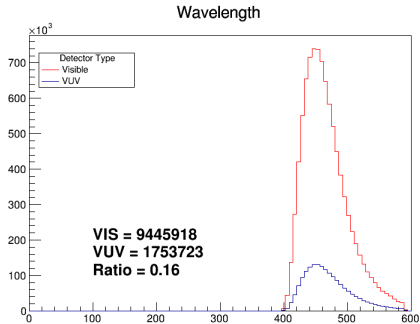
Alpha Source



Alpha Source


$$LY \approx 11000 \text{ phel/alpha}$$

Alpha Source



Left: Source at $z=0\text{cm}$. Right: Source at $z=8\text{cm}$.

① Main Simulation

② Prototype

③ Conclusions

④ References

Conclusions

- New FC design and different LDU arrangements showed no improvement in the LY.
- It is possible to achieve APEX performance by adding more LDUs or improving PEN efficiency.
- The prototype is implemented using photons, alpha particles. The implementation of the full alloy is being tested.
- The veto works well with photons and alpha particles.

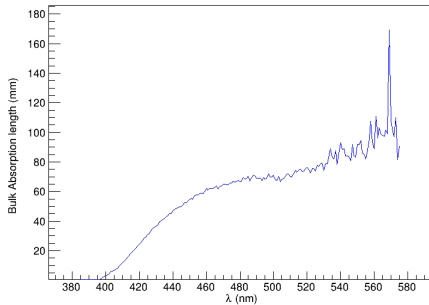
- [1] Manzanillas L. et al. *Optical properties of low background PEN structural components for the Legend-200 experiment*, **JINST**, 17 P09007 (2022).
- [2] Abraham Y. et al., *Wavelength-Shifting Performance of Polyethylene Naphthalate Films in a Liquid Argon Environment*, 2103.03232.
- [3] Araujo G. R. et al., *R&D of Wavelength-Shifting Reflectors and Characterization of the Quantum Efficiency of Tetraphenyl Butadiene and Polyethylene Naphthalate in Liquid Argon*, **Eur. Phys. J. C** 82, 442 (2022).
- [4] Boulay, M.G. et al. *Direct comparison of PEN and TPB wavelength shifters in a liquid argon detector*. **Eur. Phys. J. C** 81, 1099 (2021).

Thanks!

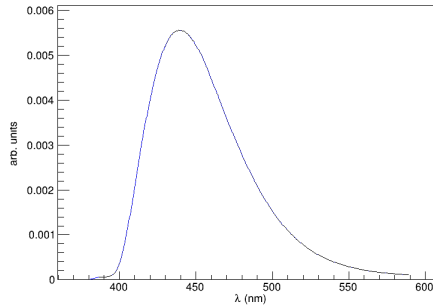
Backup slides

PolyEtylene Naphtalate - PEN

Bulk Absorption Length for PEN



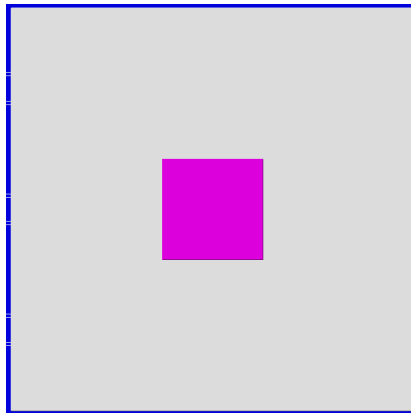
Emission spectra for PEN



Refractive Index: 1.65
Data from Luis Manzanillas.

LDUs Geometry

8x8 S14161	8x8 S14161	8x8 S14161	8x8 S14161
8x8 S14161	8x8 VUV4	8x8 VUV4	8x8 S14161
8x8 S14161	8x8 VUV4	8x8 VUV4	8x8 S14161
8x8 S14161	8x8 S14161	8x8 S14161	8x8 S14161



$$\text{Length UV} / \text{Length LDU} = 1/4$$

Reflectors

- 98% reflectance on the cathode, field cage, and membrane.
- 30% reflectance on anodes.
- Average reflectivity: 74%.

FD2 VD

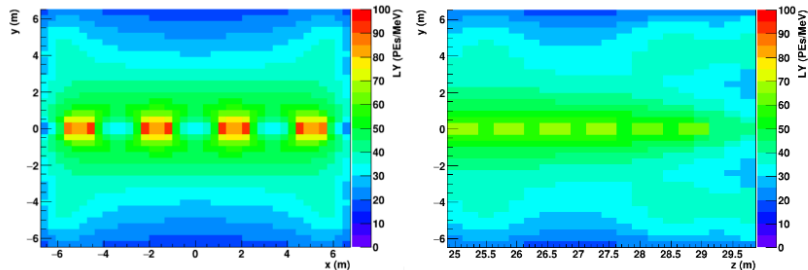


Figure 6.4: (Left) Map of the light yield (LY) in the central (x, y) transverse plane at $z = 0$ for the reference configuration. (Right) Map of LY in the central (z, y) longitudinal plane at $x = 0$, close to the detector boundary at $z = 30$ m in front of one of the membrane short walls, for the same PDS configuration.

FD2 VD

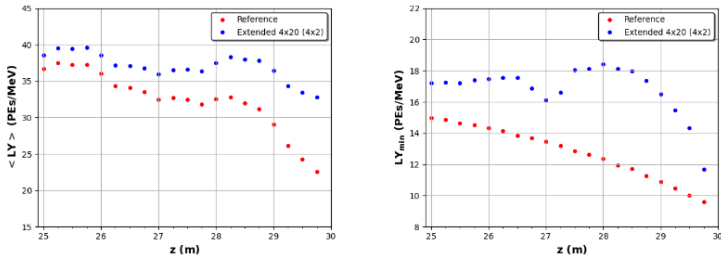


Figure 6.5: Values for average (left) and minimum (right) LY in (x, y) planes as a function of the z direction, obtained close to the detector boundary at $z = 30$ m. Light yield values for two detector configurations are shown: reference configuration including X-ARAPUCAs on the membrane short walls in blue, without short wall X-ARAPUCAs in red.

APEX

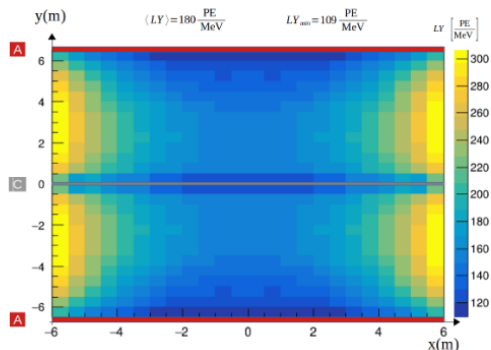
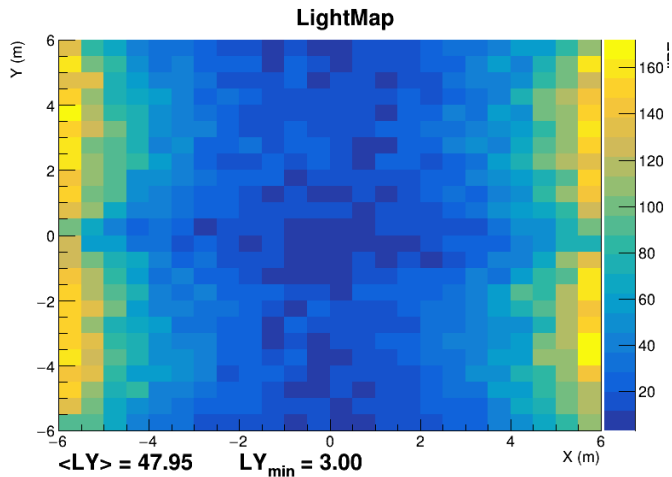
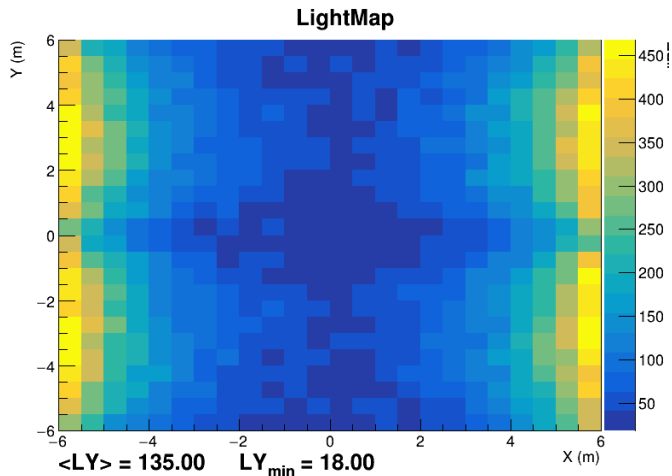


Figure 15: Map showing the expected LY in the central (x, y) transverse plane at $z = 0$ for the field cage-extended coverage APEX photon detection system. Dimmer regions are present near the anode planes (with no PDs on them) and at mid-height (near the non-instrumented cathode plane).

Pure Ar (50% PEN efficiency)



Pure Ar (100% PEN efficiency)

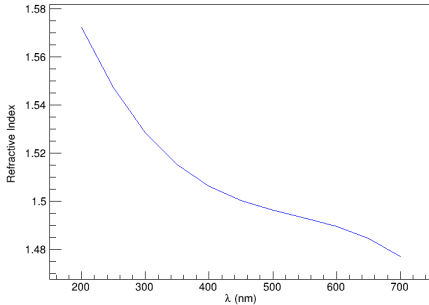


Acrylic Bulk Absorption Length

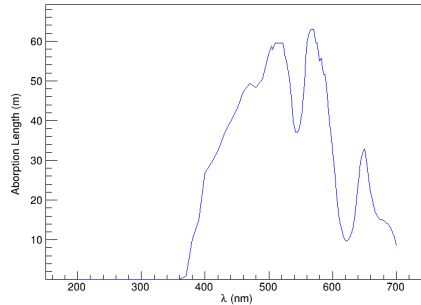
- We are using a 1mm thick acrylic foil.
- If the bulk absorption length is non-zero, we can expect some level of conversion by PEN in the case of events occurring in the buffer zone.

Poly(methyl methacrylate) - PMMA

Acrylic Refractive Index

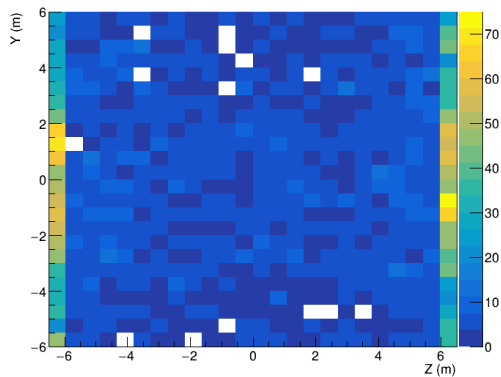
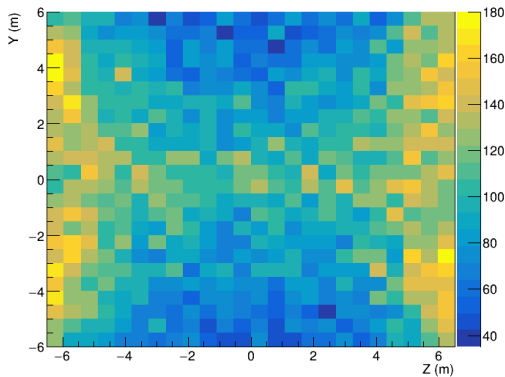


Acrylic Bulk Absorption Length



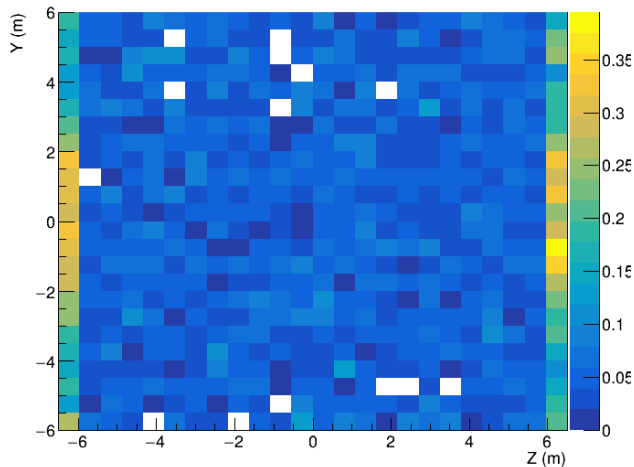
Common acrylic.
Data from Luis Manzanillas.

Lightmaps from nonideal acrylic (1cm Bulk Absorption Length)



Left: Visible. Right: VUV.

Veto from nonideal acrylic (1cm Bulk Absorption Length)



New LDU Arrays

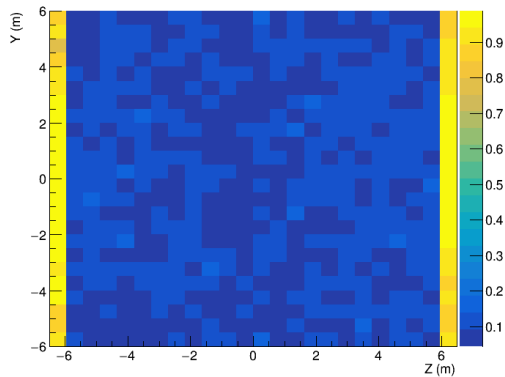
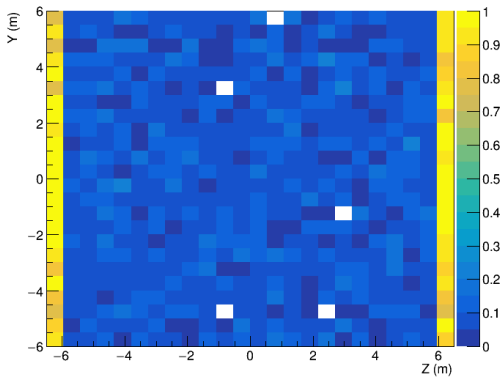
- In addition to increasing the number of LDUs we tested a new arrangement.
- In this design, the LDUs are arranged only in the 70% region.
- In the other regions, we use only VUV SiPMs.

New LDU Arrays

Variation of the space between LDUs (and the number of them). LY given in PE/MeV.
APEX: $\langle LY \rangle = 180$, $LY_{min} = 109$.

Distance	LDUs	VUV SiPMs	Area (m ²)	$\langle LY \rangle$	LYmin
1 m	1152	720	47.88	54.4	20
0.9 m	1600	800	66	74.1	34
0.8 m	1800	1260	75	82.5	38
0.7 m	2448	1428	101	113.2	57
0.6 m	2880	2160	121	133.6	58
0.5 m	4672	3212	195	193.4	97

New LDU Arrays - Veto



Left: 1m distance between LDUs. Right: 50 cm distance between LDUs.

Prototype

$$CollectionEff = \frac{Coverage}{1 - R(1 - Coverage)} = \frac{0,043}{1 - 0,98(1 - 0,043)} = 0,69$$

$$LY = 40 \times 0,5 \times 0,69 \times 0,5 \approx 7 \text{ phel/KeV}$$